

Adam Ingah

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Skills

Programming: C, Assembly, Java, Python, Verilog, HTML/CSS
Hardware: ESP32, Arduino, Raspberry Pi
Software: LTSpice, Altium, Git, SolidWorks, Fusion 360, MATLAB,

Projects

Autonomous Fire Extinguishing Robot

- Senior design capstone project where my team built an autonomous mobile robot capable of putting fires out using thermal vision on a Raspberry Pi 4
- Was responsible for hardware integration and mechanical design of rapid prototyping (3D Printed) components on the robot.

IoT Plant Irrigation System

- ESP32 Based Plant Monitoring/Watering System that uses the internet to monitor soil moisture and automatically water the plant with a pump
- Uses Arduino IoT Cloud and tracks soil moisture sensors to collect data
- Interfaced analog sensors and actuators via UART/SPI/I2C protocols

Low Vision Magnifier

- Text Magnifier with 3D printable components for individuals with low vision disabilities using Raspberry Pi, OpenCV, Tesseract, and a webcam

Four-Bit Adder PCB

- Worked and assembled a PCB in designed in Altium Designer that functioned exactly like a four-bit adder, with LED outputs and switch inputs.

Experience

Field Engineer Intern | Magellan Midstream | 5/2022 - 8/2022

- Performed petroleum sampling and lab testing
- Performed site inspections to ensure that the terminal was compliant with federal standards

Hardware Lead | UAB Engineering Lab | 8/2023 - 5/2024

- Made an economical alternative to expensive Low Vision Magnifiers for the visually impaired
- Led hardware testing and iterative prototyping in partnership with engineers and patients

Lab Assistant | UAB Libraries | 8/2022 - 5/2025

- Supported a 3D printing lab based in the library by maintaining printers and assisting patrons
- Worked closely with student engineers and researchers by having design reviews

Education

University of Washington, Seattle	M.S. Electrical Engineering	June 2027
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- GPA: 3.75 | (completing part-time, expected graduation in June 2027)
- Relevant Coursework: Quantum Computers, Digital Design with FPGAs

University of Alabama, Birmingham	B.S. Electrical Engineering	May 2025
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- GPA: 3.65/4.00 | Dean's List (Four Semesters)
- Relevant Coursework: Machine Learning, Internet of Things, Internet App Development